



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.AT.7

# Identify Equivalent Algebraic Expressions

Lesson 6-6 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can show that two expressions are equivalent by simplifying and combining like terms.

**Language Objective:** I can explain my work using the words equivalent, simplify, like terms, and combine.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent Expressions

Equivalent

Simplify

Like Terms

Combine

Coefficient



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Expressions with the same value for  allowed number are equivalent.

2

Expressions that always have the same value are .

3

To write an expression in its shortest equal form is to  it.

4

Terms with the same variable raised to the same power, like  $4x$  and  $9x$ , are .

5 To add or subtract like terms into one term is to  
them.

6 A number multiplied by a variable is a  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

JUSTIFY

**Are these expressions equivalent? How can you prove it?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Equivalent Expressions** — Expressions that look different but have the same value for every allowed variable value.

*Expresiones equivalentes — Expresiones que se ven diferentes pero tienen el mismo valor para cada valor permitido de la variable.*

- ☐ **Equivalent** — Expressions that always have the same value.

*Equivalente — Expresiones que siempre tienen el mismo valor.*

- ☐ **Simplify** — To write an expression in a shorter, simpler way.

*Simplificar — Escribir una expresión de forma más corta y simple.*

- ☐ **Like Terms** — Terms with the same letter raised to the same power, like  $2x$  and  $5x$ .

*Términos semejantes — Términos con la misma letra elevada a la misma potencia, como  $2x$  y  $5x$ .*

- ☐ **Combine** — To add or subtract terms with the same letter into one.

*Combinar — Sumar o restar términos con la misma letra en uno solo.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The expressions \_\_\_\_ equivalent because when I simplify  $4h + 2h + 30$ , I get \_\_\_\_\_. I can prove it by substituting  $h =$  \_\_\_\_: both expressions equal \_\_\_\_\_.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:**  $3x + 2x$  combines to  $5x$ , so the two forms are equivalent.

**Evidence:** Students explain  $3x + 2x = 5x$  (combining like terms), so both equal  $5x + 10$ , and verify by substituting a value like  $x = 4$  to get 30 both ways.

**Reasoning:** This evidence connects the definition of equivalent to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The expressions \_\_\_\_ equivalent because when I simplify  $4h + 2h + 30$ , I get \_\_\_\_\_. I can prove it by substituting  $h =$  \_\_\_\_: both expressions equal \_\_\_\_\_.
- My answer is \_\_\_\_\_.

*Mi respuesta es \_\_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_\_.
- I know because \_\_\_\_\_.

*Mi afirmación es \_\_\_\_\_.*

*Lo sé porque \_\_\_\_\_.*

- So \_\_\_\_\_.

*Entonces \_\_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Expressions
2. **Notes 2:** Equivalent
3. **Notes 3:** Simplify
4. **Notes 4:** Like Terms
5. **Notes 5:** Combine
6. **Notes 6:** Coefficient
7. **Watch for this mistake:** *A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms — writing  $3x + 2x$  as  $5x^2$  instead of  $5x$ . Remember:  $3x$  means 3 groups of  $x$  and  $2x$  means 2 more groups of  $x$ , so you add the coefficients ( $3 + 2 = 5$ ) and keep the variable's power the same ( $x^1$ , not  $x^2$ ).*
8. **Try It 1:** A.  $4x - 3x + x$  is 3 groups of  $x$  plus 1 more group of  $x = 4x$ . Add the coefficients:  $3 + 1 = 4$ . Check  $x = 3$ :  $3(3) + 3 = 12$  and  $4(3) = 12$ .
9. **Try It 2:** A.  $2x + 5$  — Order of addition does not change the value (commutative property):  $5 + 2x = 2x + 5$ . Check  $x = 10$ :  $5 + 2(10) = 25$  and  $2(10) + 5 = 25$ .